

IRIS FLOWER CLASSIFICATION

CODSOFT DATA SCIENCE INTERNSHIP – TASK 3

OBJECTIVE

The objective of this project is to develop a Machine Learning model that classifies Iris flowers into three species: Iris-setosa, Iris-versicolor, and Iris-virginica based on sepal and petal measurements. This project helps in understanding classification algorithms and predictive modeling.

PROJECT DESCRIPTION

The Iris Flower Classification project includes:

-  Loading the Iris dataset
-  Data preprocessing and preparation
-  Training a Machine Learning classification model
-  Predicting flower species
-  Evaluating model performance using accuracy

TOOLS USED

 Python

 VS Code

 Pandas

 Scikit-learn

 NumPy

FEATURES

1. Dataset Loading

Loads the Iris dataset containing flower measurements.

2. Data Preprocessing

Prepares the dataset for Machine Learning.

3. Model Training

Uses Logistic Regression to train the classification model.

4. Species Prediction

Predicts the species of Iris flowers.

5. Performance Evaluation

Measures model accuracy on test data.

DATASET INFORMATION

Dataset Name 

iris.csv

Features

- ✓ Sepal Length
- ✓ Sepal Width
- ✓ Petal Length
- ✓ Petal Width

Target Classes

- ✓ Iris-setosa
- ✓ Iris-versicolor
- ✓ Iris-virginica

CONCEPTS USED

- ✓ Machine Learning
- ✓ Classification
- ✓ Logistic Regression
- ✓ Train-Test Split
- ✓ Accuracy Score
- ✓ Data Analysis

KEY LEARNINGS

- ✓ Understanding classification algorithms
- ✓ Working with real-world datasets
- ✓ Training and testing Machine Learning models
- ✓ Evaluating model performance
- ✓ Predicting categories using data

INSIGHTS

Machine Learning can accurately classify flower species using measurements.

Logistic Regression is effective for multi-class classification problems.

Feature values play an important role in determining flower species.

The Iris dataset is one of the most popular datasets for learning Machine Learning concepts.

CONCLUSION

The Iris Flower Classification project successfully demonstrates how Machine Learning can classify flowers into different species based on their measurements. The Logistic Regression model achieved 100% accuracy, showing excellent classification performance on the Iris dataset.

OUTPUT

```
Iris Flower Classification Completed Successfully!  
Accuracy: 100.0 %
```

- ✓ Iris Dataset Loaded Successfully
- ✓ Model Training Completed
- ✓ Flower Species Predicted Successfully
- ✓ Accuracy Calculated
- ✓ Accuracy Achieved: 100.0%

  AUTHOR

Chinta Ramalakshmi

 CodSoft Data Science Intern